What is Machine Learning?

A computer program is said to learn if its performance at task **T**, as measured by **P**, improves with experience **E**.

- Tom Mitchell

e.g., Google Maps finding the fastest route based on current traffic?

Google Maps predicting traffic jams based on historical traffic patterns?

millions of past trips from other users

Experience (**E**)

traffic jam prediction

Task (**T**)

Prediction accuracy

Performance Measure (**P**)

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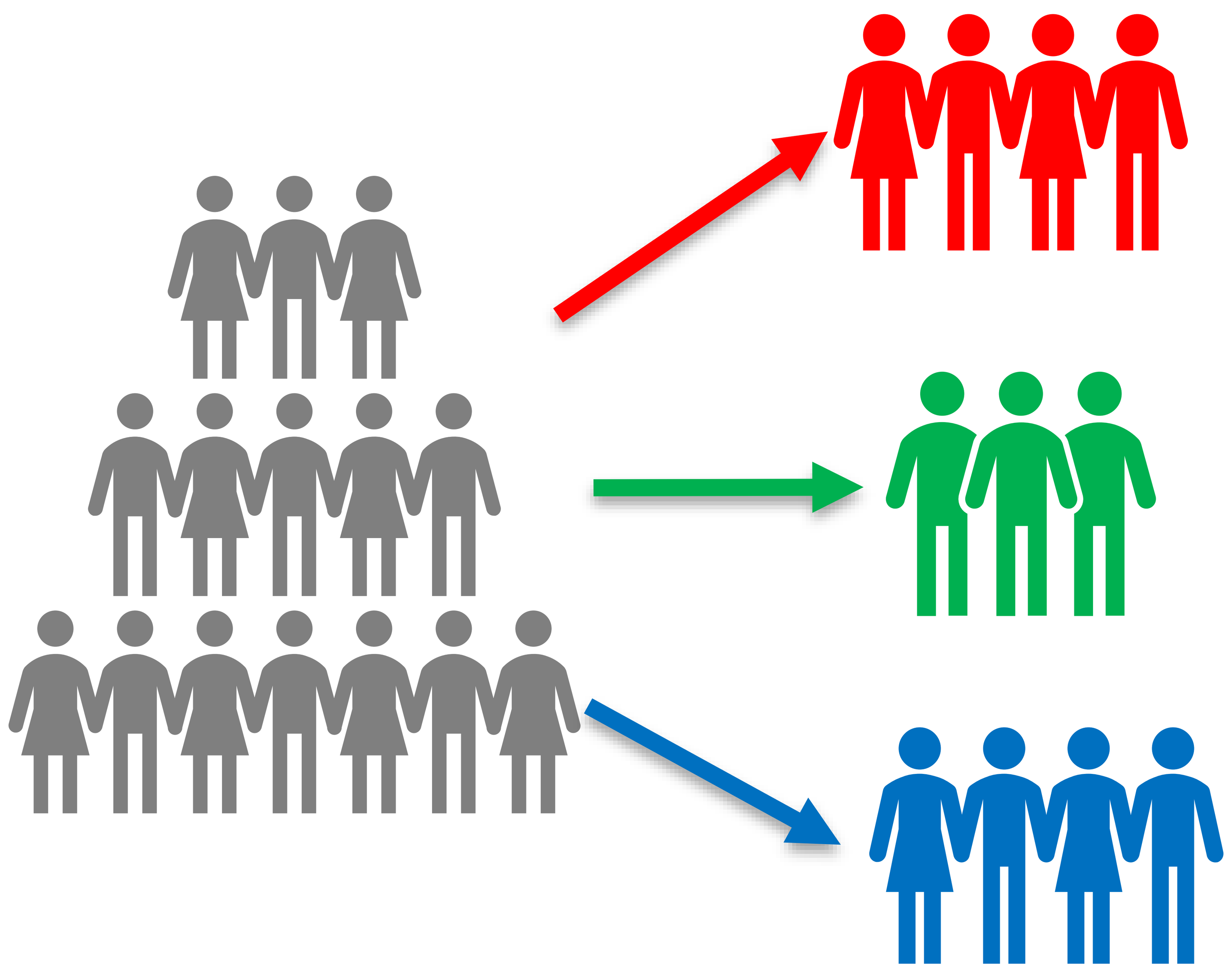
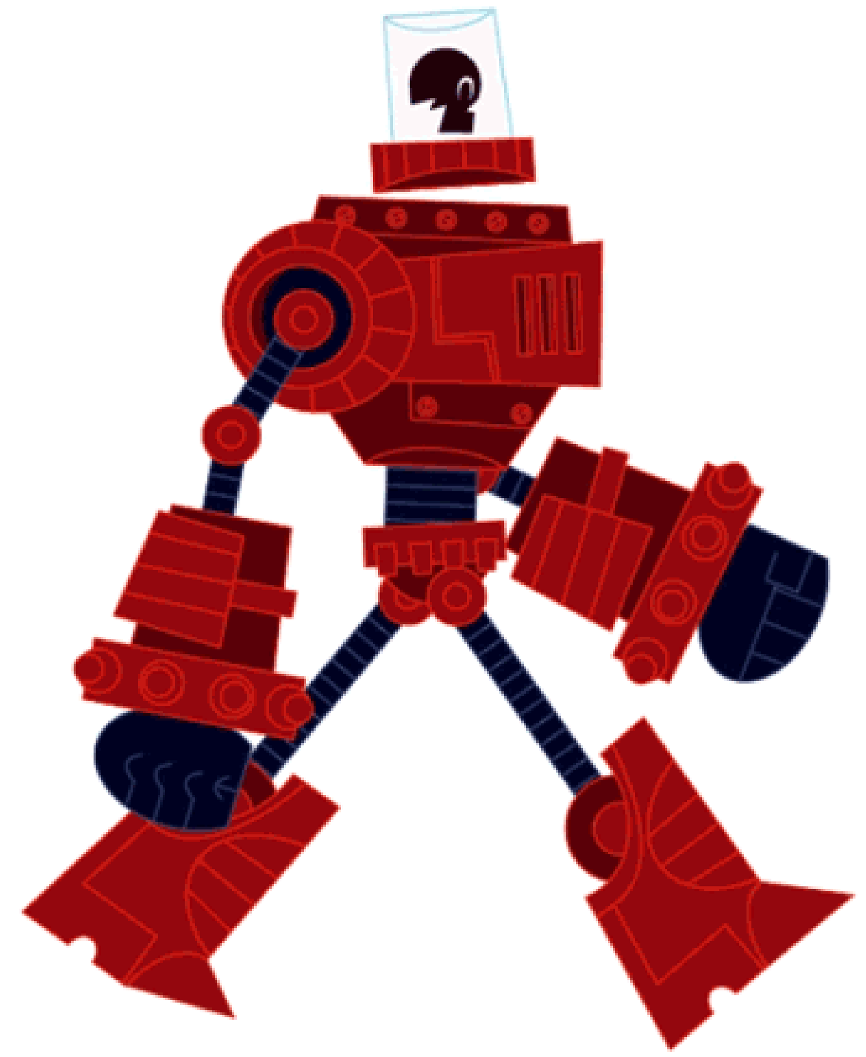
What is Machine Learning?

Examples: Task (**T**) Experience (**E**) Performance (**P**)





T: Spam Email Detection
E: Email Data
P: Detection Accuracy

T: Customer Segmentation
E: Purchase data
P: Silhouette Scores

T: Robot Walking
E: Past records of Walking
P: Walking time

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Types of Machine Learning

-Supervised Learning

-Unsupervised Learning

-Reinforcement Learning

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Types of Machine Learning

➢ Supervised Learning

Input: data with desired output

Output: A function(i.e., model) that maps data to the desired output

Data	desired output
	Apple
⋮	⋮
	Pear

→ $\boxed{\text{Supervised Learning}}$ → $f(\cdot)$ s.t.,

$f($  $) =$ Apple

$f($  $) =$ Pear

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Types of Machine Learning

➢ Supervised Learning

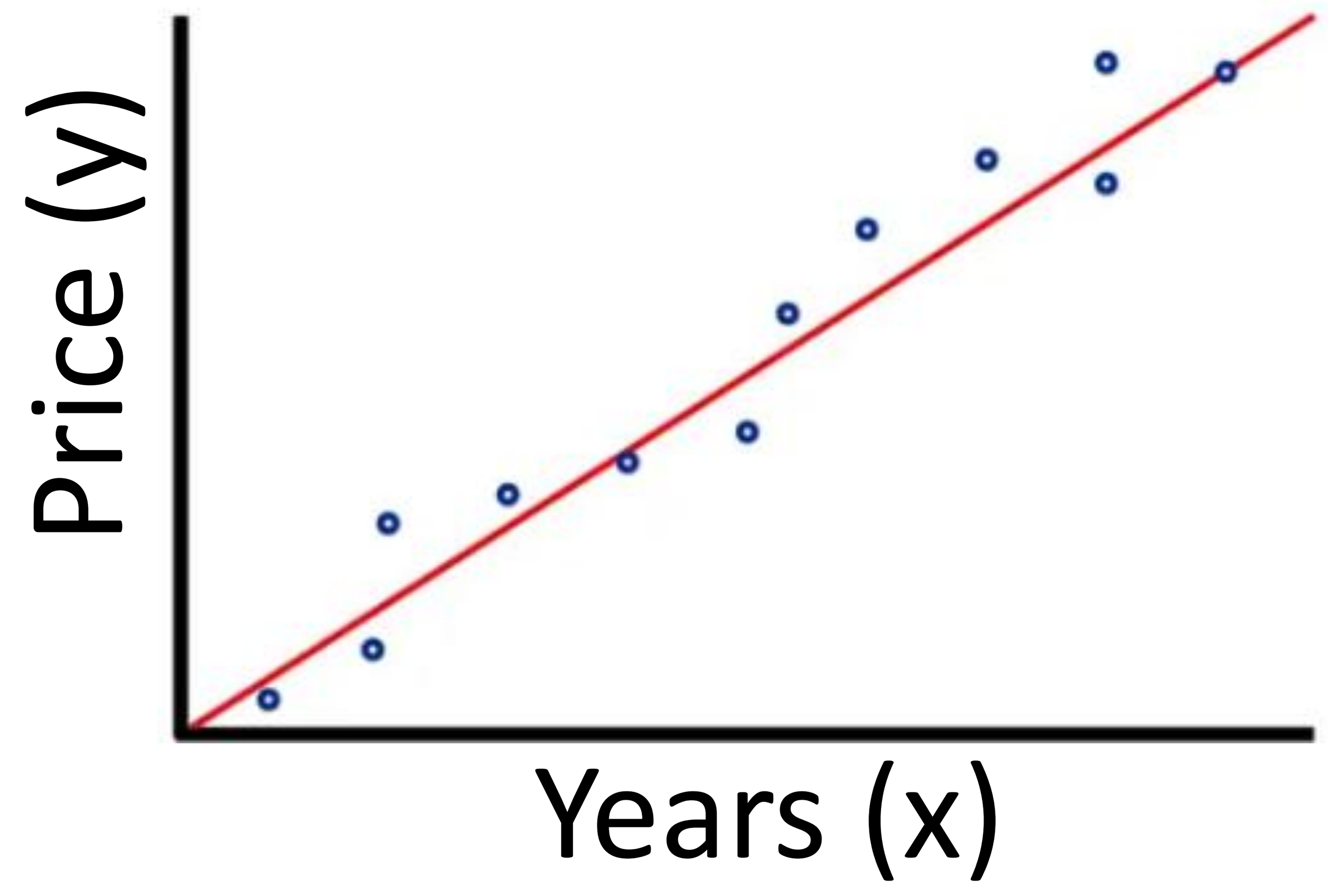
Input: data with desired output

Output: A function(i.e., model) that maps data to the desired output

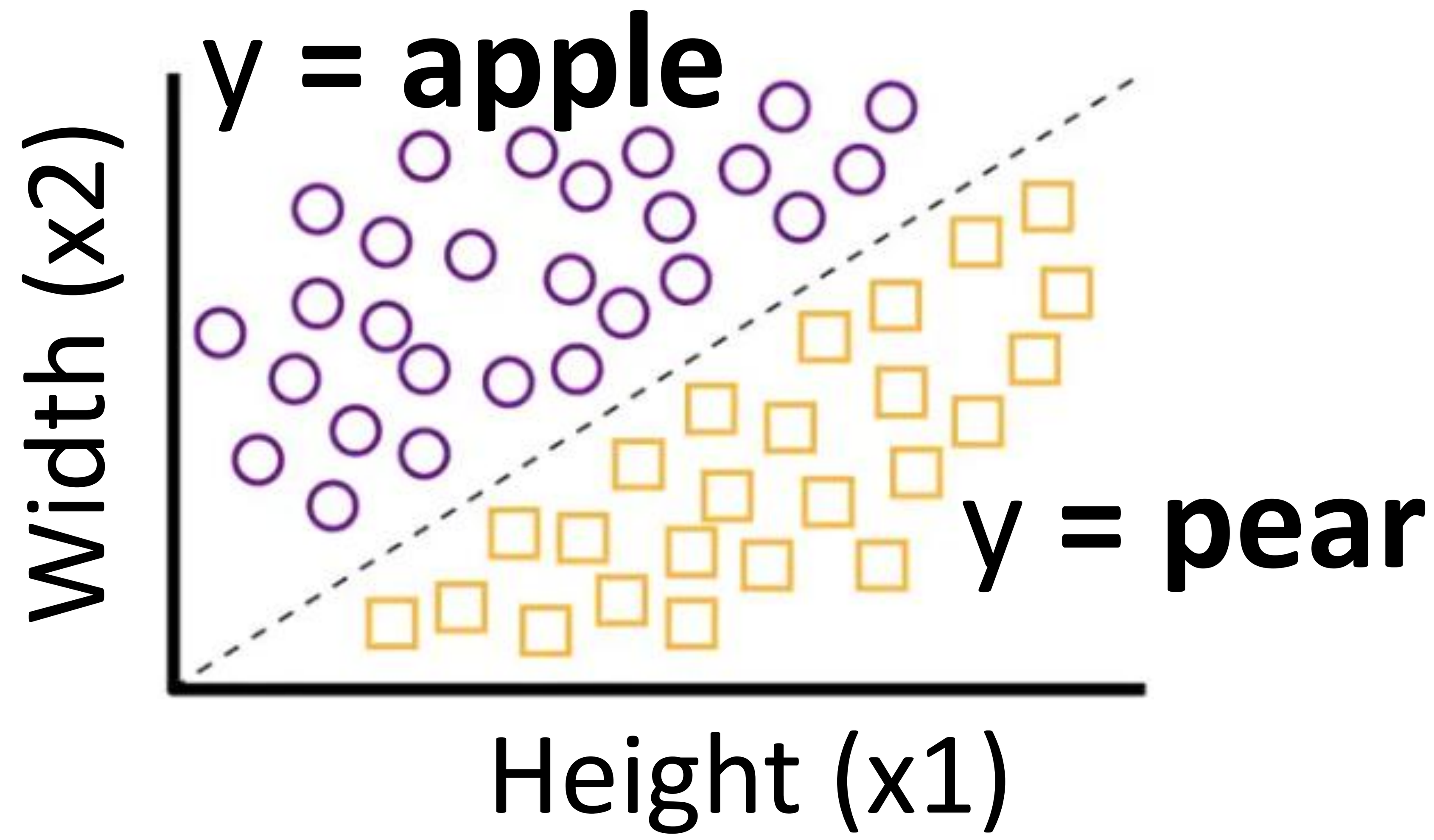
Desired output

Numerical → Regression **Lecture 8**

Categorical → Classification **Lecture 9**



find the line that best aligns with samples



find the line that separates two classes

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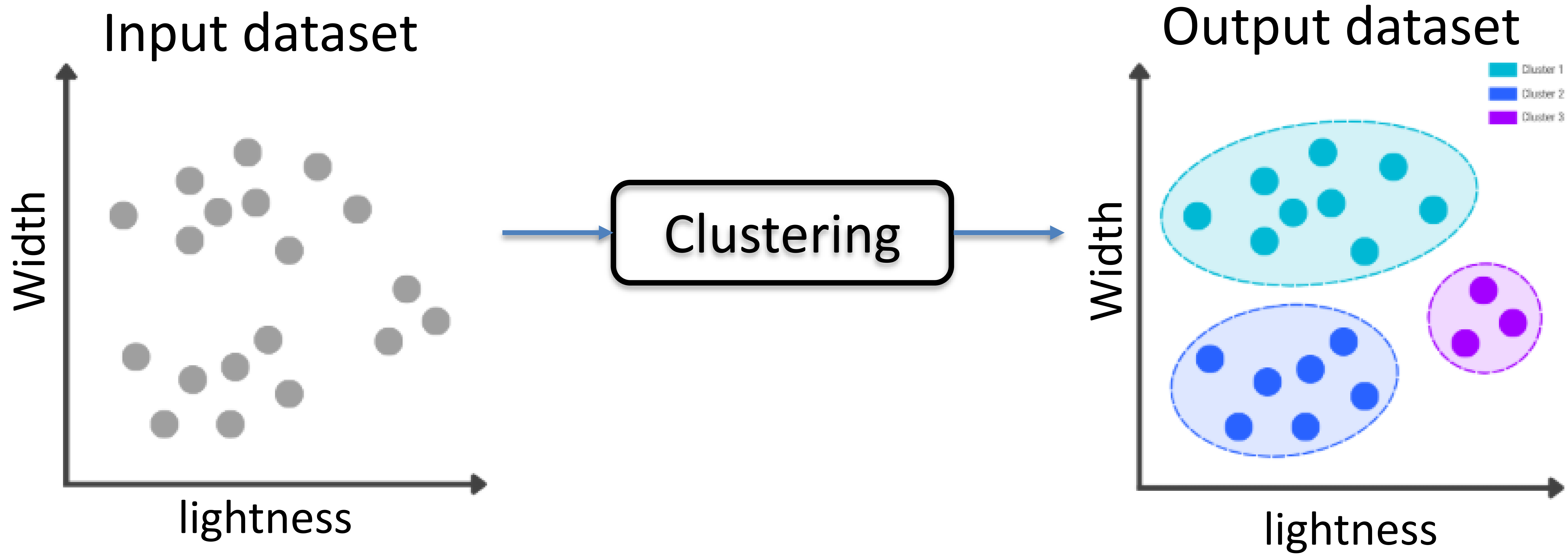
Types of Machine Learning

➢ Unsupervised Learning

Input: data (No desired output)

Output: Underlying patterns in data

Clustering **Lecture 10**



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Types of Machine Learning

➢ Reinforcement Learning

Input: Sequence of States, Actions, and Delayed Rewards

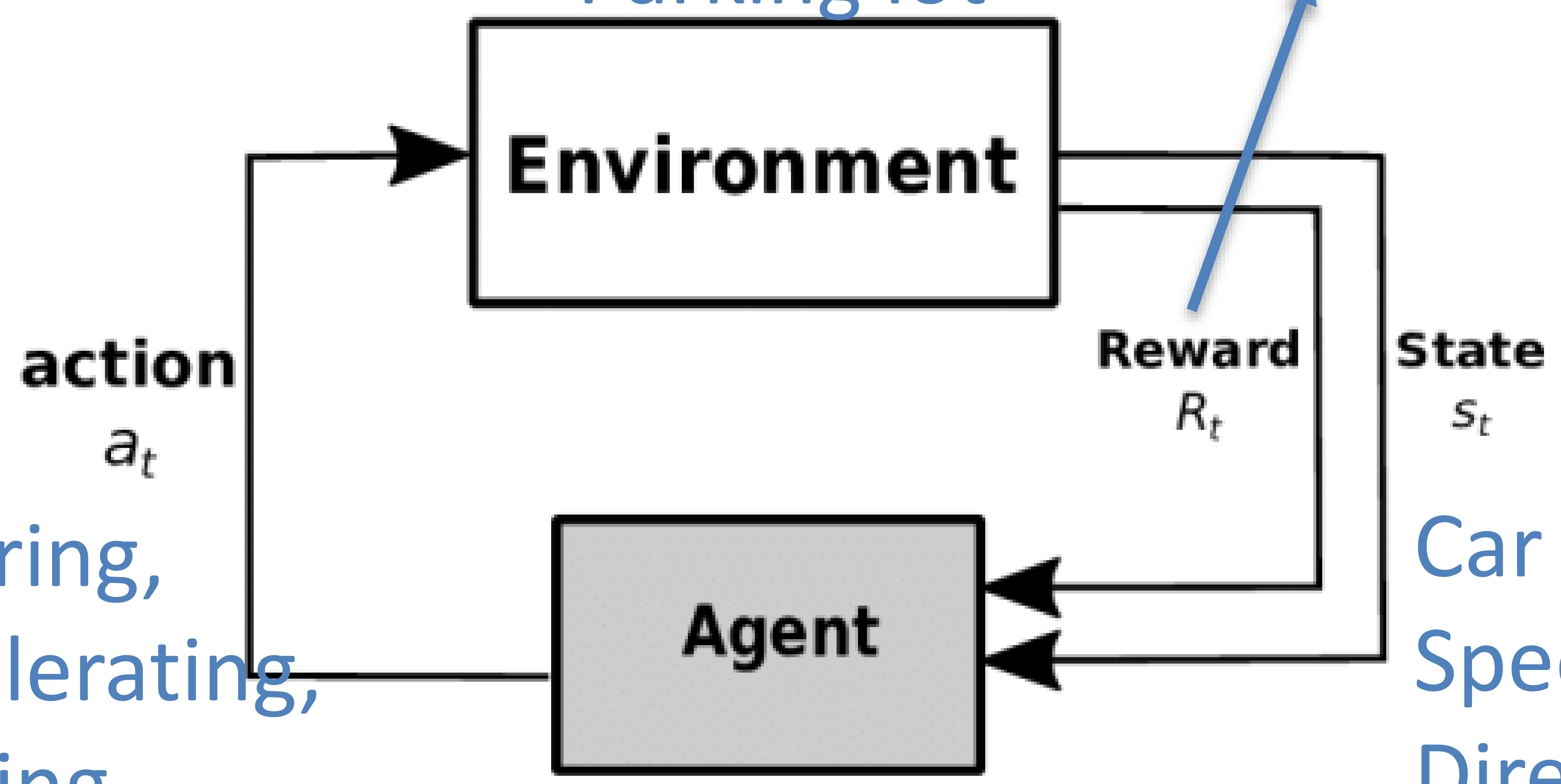
Learn how to decide

Output: Action Strategy (a rule that maps the environment to action)

Positive: closer to the correct spot

Negative: hits obstacles or parks badly

Parking lot



Steering,
Accelerating,
Braking,
…

Self-driving car

Car location,
Speed,
Direction,
Obstacle locations

Self-driving car learn to park

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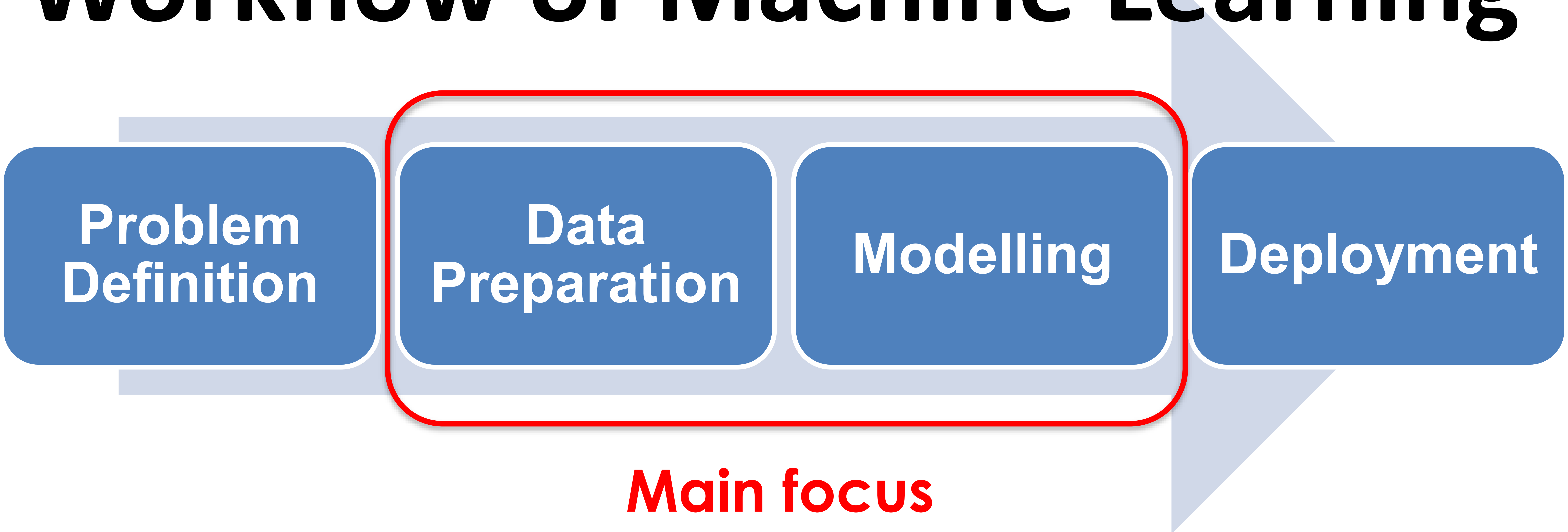
Workflow of Machine Learning



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Workflow of Machine Learning

Problem Definition → Data Preparation → Modelling → Deployment

	Example
1. Clarify the objective	
What problem are you trying to solve?	Predict House Prices
2. Determine if ML is appropriate	
Can this problem be learned from data?	Yes
3. Specify inputs and outputs	
What input features will the model use?	Size, location, no. rooms, etc.
What is the model trying to predict?	Sale price in SGD
4. Choose the task type	
Classification? Regression? Clustering?...	Regression
5. Decide on performance measure	
Accuracy? Mean Squared Error?...	Mean Squared Error

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Workflow of Machine Learning

Problem Definition → **Data Preparation** → Modelling → Deployment

1. Data Collection

Gather raw data from sensors, databases, files, etc.

e.g., Predict House Prices

No.	Size	No. rooms	Location	Sold Price
1				
…		Features		Desired output
100				

2. Exploratory Data Analysis

Flag any anomalies and inconsistencies.

No.	Size	No. rooms	Location	Sold Price
1	112	2	“Pioneer”	600000
2		4	“Clementi"	801256
3	100	3	4	“650K”
4	116	3	“City Hall”	2300000
…				
100	112	2	“Pioneer”	600000

Missing value

Inconsistent data values

Outlier

Duplicate data

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Workflow of Machine Learning

Problem Definition | **Data Preparation** | Modelling | Deployment

3. Data Preprocessing

- Address the problems flagged in the previous step.

  e.g., handling missing value through **dropping** or **imputation**

If the number of samples is large.

Option 1: take the average value of the corresponding feature.

Option 2: take a value outside the normal range.

Missing value

No.	Size	No. rooms	Location	Sold Price
1	112	2	“Pioneer”	600000
2		4	“Clementi"	801256
3	100	3	4	“650K”
4	116	3	“City Hall”	1300000
…				
100	112	2	“Pioneer”	600000

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Workflow of Machine Learning

Problem Definition → **Data Preparation** → Modelling → Deployment

3. Data Preprocessing

- Format the data

  ❖ Convert non-numerical data into numerical form.

  e.g., Use postal code for location

  Convert categories into a binary vector through **one-hot encoding.**

  Categories: ['Cat', 'Dog', 'Bird'] $\xrightarrow{\textbf{one-hot encoding}}$ Cat: [1,0,0] Dog: [0,1,0] Bird: [0,0,1]

  0 1 2

  $\frac{0-(-1)}{1-(-1)}$

  ❖ **Normalization (Ensure Features Contribute Fairly)**

  Across Samples for Each Feature

  $\frac{3-2}{4-2}=\frac{1}{2}$

No.	Feature 1	Feature 2
1	3	-1
2	2 min 0	0
3	4 max 1	1

Use only training data Statistics!

Min-max normalization: $x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}}$

Z-score standardization: $x_{scaled} = \frac{x-\mu}{\sigma}$

μ → Mean

σ → Standard deviation

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Workflow of Machine Learning

Problem Definition → **Data Preparation** → Modelling → Deployment

4. Feature Selection or Feature Extraction

Manipulating existing features to obtain **more relevant** features that improves performance

Feature Selection	Feature Extraction
Chooses from **existing features**	Creates **new features** from original ones
e.g., drop “Favorite color”	e.g., combine “height” and “weight” into “BMI”

Example task: Predict health risk

	Height (m)	Weight (kg)	Age	Gender	Favorite color	Risk
1	1.75	59	33	Male	Red	High
…	…	…	…	…	…	…
100	1.65	50	31	Female	Blue	Medium

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Workflow of Machine Learning

Problem Definition | **Data Preparation** | Modelling | Deployment

4. Feature Selection and Feature Extraction

Manipulating existing features to obtain **more relevant** features that improves performance

-Feature Selection with **Pearson's correlation coefficient (r)**

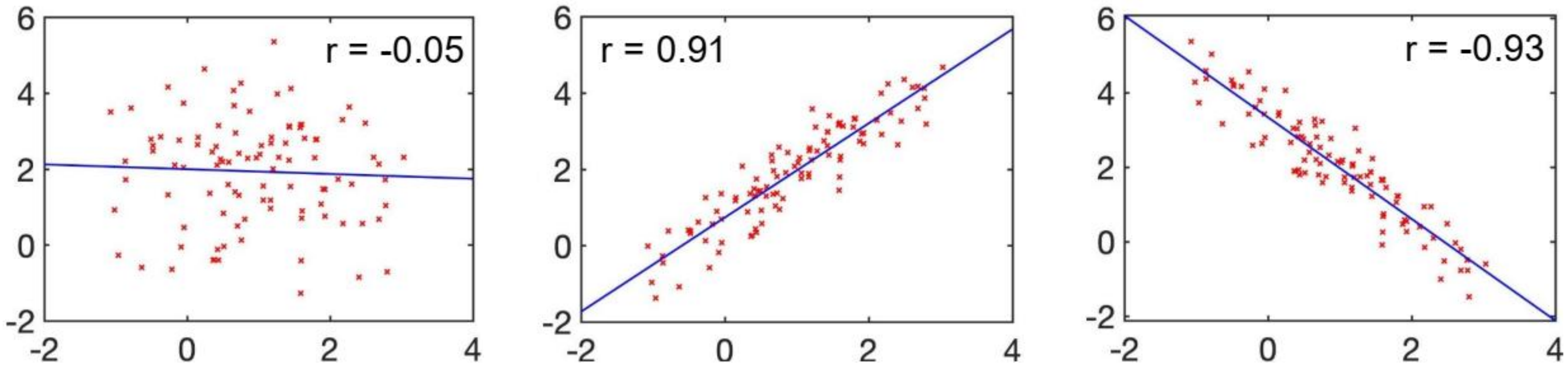
(Measures linear relationship between two numerical variables)

Given N pairs of data points $\{(a_1, b_1), \ldots, (a_N, b_N)\}$

Use only training data!

$$r = \frac{\frac{1}{N}\sum_{i=1}^{N}(a_i - \overline{a})(b_i - \overline{b})}{\sqrt{\frac{1}{N}\sum_{i=1}^{N}(a_i - \overline{a})^2}\sqrt{\frac{1}{N}\sum_{i=1}^{N}(b_i - \overline{b})^2}}$$

where $\overline{a} = \frac{1}{N}\sum_{i=1}^{N} a_i$, $\overline{b} = \frac{1}{N}\sum_{i=1}^{N} b_i$



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Workflow of Machine Learning

Problem Definition | **Data Preparation** | Modelling | Deployment

4. Feature Selection and Feature Extraction

Manipulating existing features to obtain **more relevant** features that improves performance

-Feature Selection with **Pearson's correlation coefficient (r)**

(Measures linear relationship between two numerical variables)

Way 1:
- Compute r between each input feature and the target output in the **training set**.
- Select features with high absolute r.

Regression

Regression / Classification

Way 2:
- Compute r between every pair of input features in the **training set**
- Remove features with high absolute r.

Duplicate feature (Redundant)

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Workflow of Machine Learning

Problem Definition | **Data Preparation** | Modelling | Deployment

5. Split data into training, validation and test sets

Training	Validation	Test
For training the ML models (learn internal parameters)	For choosing the hyperparameter of the model	For testing the "real" performance and generalization

Seen/visible data

Unseen/hidden data

No.	Size	No. rooms	Location	Sold Price
1				
…		Training Set		
80				

No.	Size	No. rooms	Location	Sold Price
81				
…		Validation Set		
90				

No.	Size	No. rooms	Location	Sold Price
91				
…		Test Set		
100				

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Workflow of Machine Learning

Problem Definition | Data Preparation | **Modelling** | Deployment

1. Train model on the training set

	No.	Size	No. rooms	Location	Sold Price
$\boldsymbol{x}_1$	1	$x_{1,1}$	$x_{1,2}$	$x_{1,3}$	y_1
	…				
$\boldsymbol{x}_{80}$	80	$x_{80,1}$	$x_{80,2}$	$x_{80,3}$	y_{80}

$\{(\boldsymbol{x}_1, y_1), (\boldsymbol{x}_2, y_2), \dots, (\boldsymbol{x}_{80}, y_{80})\}$

→ Learning Algorithm → Learned Model $f(\cdot): \boldsymbol{X} \rightarrow \boldsymbol{y}$

2. Evaluate the model on the validation set

	No.	Size	No. rooms	Location	Sold Price
$\boldsymbol{x}_{81}$	81	$x_{81,1}$	$x_{81,2}$	$x_{81,3}$	y_{81}
	…				
$\boldsymbol{x}_{90}$	90	$x_{90,1}$	$x_{90,2}$	$x_{90,3}$	y_{90}

→ Learned Model $f(\cdot)$ → Predicted Prices: $y'_{81}, y'_{82}, \dots, y'_{90}$

Performance measure P

Workflow of Machine Learning

Problem Definition | Data Preparation | **Modelling** | Deployment

3. Hyperparameter Tuning

Change the hyperparameters and repeat steps 1&2 to find the best hyperparameters that have the best validation performance.

	Hyperparameters	Parameters
Definition	External settings chosen before training	Internal variables learned from the data
Learned during the training?	No	Yes
Set by user?	Yes	No
Examples	-Learning rate -Training steps	-Regression Coefficents in Linear Regression

4. Test model on the test set

Unbiased evaluation of the final model

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Workflow of Machine Learning

Problem Definition → Data Preparation → Modelling → **Deployment**

- **Model Deployment (real-world environment)**

  Web API : Model exposed as an endpoint that APPs can call

  Cloud: AWS, GCP, or Azure provide managed ML serving platforms.

  Edge: Model runs locally on devices like smartphones, IoT devices, etc.

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Summary by Quick Quiz

- ✓ Three Components in ML Definition
  Task, performence measure, experience
- ✓ Three Types of ML
- ✓ Three ways of Handling Missing Value
  Dropping, take average, choosing a value
- ✓ Data Formatting out of range
  Categorical data? Different Scales for Features?
- ✓ One Criterion for Feature Selection
- ✓ Dataset Partition
  Training, validating, testing

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Review of Fundamentals

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Review of Fundamentals

❖ Scalars, Vectors, Matrices

- A **scalar** is a simple numerical value (e.g., 10, -4.2,).

  Denoted by an *italic* letter (e.g., x, y)

- A **vector** is an ordered list of scalar values.

  Denoted by a **bold** letter (e.g., $\mathbf{x}$, $\mathbf{y}$)

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- A **matrix** is a rectangular array of numbers arranged in rows and columns

  Denoted by a **bold** CAPITAL letter (e.g., $\mathbf{X}$, $\mathbf{Y}$)

$$\mathbf{X} = \begin{bmatrix} x_{1,1} & x_{1,2} & x_{1,3} \\ x_{2,1} & x_{2,2} & x_{2,3} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1.5 \\ 2 & 9 & -3.2 \end{bmatrix}$$

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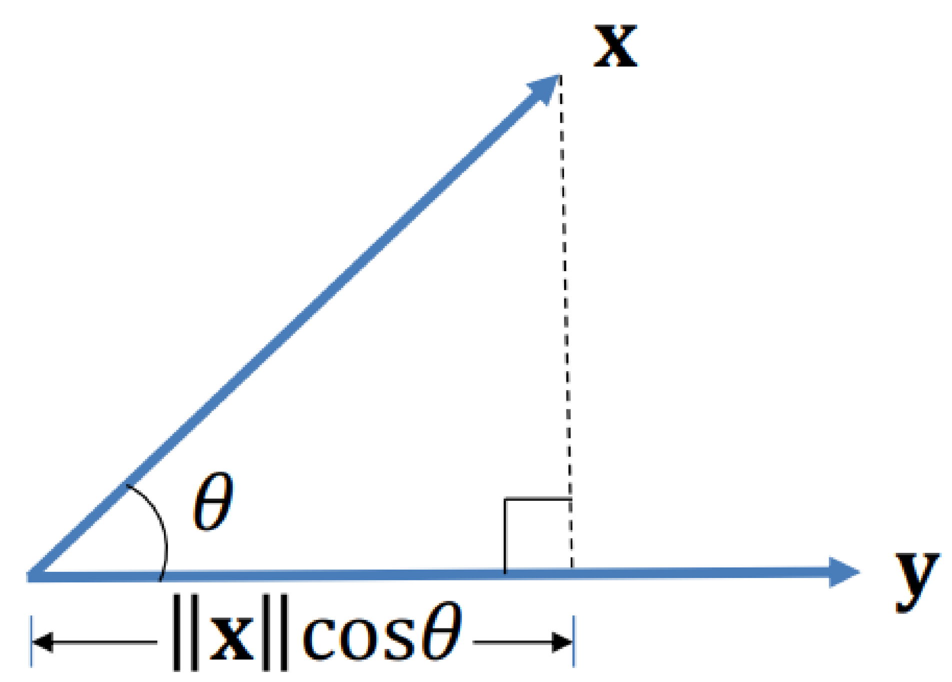
Review of Fundamentals

❖ Operations on Vectors and Matrices

$$\mathbf{x}+\mathbf{y}=\begin{bmatrix}x_1\\x_2\end{bmatrix}+\begin{bmatrix}y_1\\y_2\end{bmatrix}=\begin{bmatrix}x_1+y_1\\x_2+y_2\end{bmatrix}$$

$$\mathbf{x}=\begin{bmatrix}x_1\\x_2\end{bmatrix},\ \mathbf{x}^T=[x_1\ \ x_2]$$

$$\mathbf{x}-\mathbf{y}=\begin{bmatrix}x_1\\x_2\end{bmatrix}-\begin{bmatrix}y_1\\y_2\end{bmatrix}=\begin{bmatrix}x_1-y_1\\x_2-y_2\end{bmatrix}$$



$$\mathbf{x}\cdot\mathbf{y}=\mathbf{x}^T\mathbf{y}=\|\mathbf{x}\|\,\|\mathbf{y}\|\cos\theta$$

$$=[x_1\ \ x_2]\begin{bmatrix}y_1\\y_2\end{bmatrix}=x_1\,y_1+x_2y_2$$

$$a\,\mathbf{x}=a\begin{bmatrix}x_1\\x_2\end{bmatrix}=\begin{bmatrix}ax_1\\ax_2\end{bmatrix}$$

$$\mathbf{Wx}=\begin{bmatrix}w_{1,1} & w_{1,2} & w_{1,3}\\ w_{2,1} & w_{2,2} & w_{2,3}\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}$$

$$=\begin{bmatrix}w_{1,1}x_1+w_{1,2}x_2+w_{1,3}x_3\\ w_{2,1}x_1+w_{2,2}x_2+w_{2,3}x_3\end{bmatrix}$$

$$\mathbf{x}^T\mathbf{W}=[x_1\ \ x_2]\begin{bmatrix}w_{1,1} & w_{1,2} & w_{1,3}\\ w_{2,1} & w_{2,2} & w_{2,3}\end{bmatrix}$$

$$=[(x_1w_{1,1}+x_2w_{2,1})\quad (x_1w_{1,2}+x_2w_{2,2})\quad (x_1w_{1,3}+x_2w_{2,3})]$$

$$\boldsymbol{A}_{3\times3}\boldsymbol{B}_{3\times3}=\begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix}\times\begin{bmatrix}b_{11} & b_{12} & b_{13}\\ b_{21} & b_{22} & b_{23}\\ b_{31} & b_{32} & b_{33}\end{bmatrix}=\begin{bmatrix}a_{11}\times b_{11}+a_{12}\times b_{21}+a_{13}\times b_{31} & a_{11}\times b_{12}+a_{12}\times b_{22}+a_{13}\times b_{32} & a_{11}\times b_{13}+a_{12}\times b_{23}+a_{13}\times b_{33}\\ a_{21}\times b_{11}+a_{22}\times b_{21}+a_{23}\times b_{31} & a_{21}\times b_{12}+a_{22}\times b_{22}+a_{23}\times b_{32} & a_{21}\times b_{13}+a_{22}\times b_{23}+a_{23}\times b_{33}\\ a_{31}\times b_{11}+a_{32}\times b_{21}+a_{33}\times b_{31} & a_{31}\times b_{12}+a_{32}\times b_{22}+a_{33}\times b_{32} & a_{31}\times b_{13}+a_{32}\times b_{23}+a_{33}\times b_{33}\end{bmatrix}$$

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Review of Fundamentals

❖ Operations on Vectors and Matrices

-**Matrix inverse**: $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})}\mathrm{adj}(\mathbf{A})$

```python
import numpy as np
# Calculate inverse
A_inv = np.linalg.inv(A)
```

- det(**A**): determinant of **A**

det(**A**)=$\sum_{j=1}^{k} a_{ij}\mathbf{C}_{ij}$, where cofactor $\mathbf{C}_{ij} = (-1)^{i+j}\mathbf{M}_{ij}$

$\mathbf{M}_{ij}$: minor of a_{ij} is the determinant obtained by deleting the row and column in which a_{ij} lies.

e.g.,

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\mathbf{M}_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22} \times a_{33} - a_{23} \times a_{32}$$

$$\mathbf{C}_{11} = (-1)^{1+1}\mathbf{M}_{11}$$

$$\det(\mathbf{A}) = \sum_{j=1}^{3} a_{ij}\mathbf{C}_{ij} = a_{i1}\mathbf{C}_{i1} + a_{i2}\mathbf{C}_{i2} + a_{i3}\mathbf{C}_{i3}$$

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Review of Fundamentals

❖ Operations on Vectors and Matrices

-**Matrix inverse**: $\mathbf{A}^{-1} = \dfrac{1}{\det(\mathbf{A})}\operatorname{adj}(\mathbf{A})$

- adj(**A**): adjugate/adjoint of **A**

adj(**A**)=$\mathbf{C}^T$, where cofactor $\mathbf{C}_{ij} = (-1)^{i+j}\mathbf{M}_{ij}$

$\mathbf{M}_{ij}$: minor of a_{ij} is the determinant obtained by deleting the row and column in which a_{ij} lies.

e.g.,

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\mathbf{M}_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22} \times a_{33} - a_{23} \times a_{32}$$

$$\mathbf{C}_{11} = (-1)^{1+1}\mathbf{M}_{11}$$

$$\text{adj}(\mathbf{A}) = \mathbf{C}^T = \begin{bmatrix} \mathbf{M}_{11} & -\mathbf{M}_{21} & \mathbf{M}_{31} \\ -\mathbf{M}_{12} & \mathbf{M}_{22} & \mathbf{M}_{32} \\ \mathbf{M}_{13} & -\mathbf{M}_{23} & \mathbf{M}_{33} \end{bmatrix}$$

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Review of Fundamentals

❖ Operations on Vectors and Matrices

-**Matrix inverse**: $\mathbf{A}^{-1} = \dfrac{1}{\det(\mathbf{A})}\mathrm{adj}(\mathbf{A})$

- det(**A**)=$\sum_{j=1}^{k} a_{ij}\mathbf{C}_{ij}$, where cofactor $\mathbf{C}_{ij} = (-1)^{i+j}\mathbf{M}_{ij}$
- adj(**A**)=$\mathbf{C}^T$

$\mathbf{M}_{ij}$: minor of a_{ij} is the determinant obtained by deleting the row and column in which a_{ij} lies.

e.g., What is the cofactor matrix of $\mathbf{A} = \begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix}$

$$\mathbf{C} = \begin{bmatrix} -3 & 6 & -3 \\ 6 & -12 & 6 \\ -3 & 6 & -3 \end{bmatrix}$$

$$\mathbf{C}_{11} = (-1)^{1+1}\begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} = -3,\ \mathbf{C}_{12} = (-1)^{1+2}\begin{vmatrix} 3 & 5 \\ 6 & 8 \end{vmatrix} = 6,\ \mathbf{C}_{13} = (-1)^{1+3}\begin{vmatrix} 3 & 4 \\ 6 & 7 \end{vmatrix} = -3$$

$$\mathbf{C}_{21} = (-1)^{2+1}\begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} = 6,\ \mathbf{C}_{22} = (-1)^{2+2}\begin{vmatrix} 0 & 2 \\ 6 & 8 \end{vmatrix} = -12, \mathbf{C}_{23} = (-1)^{2+3}\begin{vmatrix} 0 & 1 \\ 6 & 7 \end{vmatrix} = 6,$$

$$\mathbf{C}_{31} = (-1)^{3+1}\begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix} = -3,\ \mathbf{C}_{32} = (-1)^{3+2}\begin{vmatrix} 0 & 2 \\ 3 & 5 \end{vmatrix} = 6, \mathbf{C}_{33} = (-1)^{3+3}\begin{vmatrix} 0 & 1 \\ 3 & 4 \end{vmatrix} = -3,$$

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Review of Fundamentals

❖ Operations on Vectors and Matrices

-**Matrix inverse**: $\mathbf{A}^{-1} = \dfrac{1}{\det(\mathbf{A})}\operatorname{adj}(\mathbf{A})$

Matrix inverse $\mathbf{A}^{-1}$ exists if and only if the matrix $\mathbf{A}$ is **square and nonsingular** (i.e., **invertible)**

In other words, there exists a square matrix $\mathbf{B}$, such that **AB**=**BA**=**I** (Identity matrix)

$$\mathbf{I}_{d\times d} = \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & 0 & \ddots & 0 & \vdots \\ 0 & \vdots & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$$

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Review of Fundamentals

❖ Operations on Vectors and Matrices

-**Matrix inverse**: $\mathbf{A}^{-1} = \dfrac{1}{\det(\mathbf{A})}\mathrm{adj}(\mathbf{A})$

Matrix inverse $\mathbf{A}^{-1}$ exists if and only if the matrix $\mathbf{A}$ is **square and nonsingular** (i.e., **invertible)**

2 methods to check invertibility

det(**A**)=0?

rank($\mathbf{A}_{d\times d}$)=d?

rank($\mathbf{A}$): maximal number of **linearly independent** columns/rows of **A**

$\beta_1\mathbf{a}_1 + \beta_2\mathbf{a}_2 + \cdots + \beta_m\mathbf{a}_m = 0$ only holds for $\beta_1 = \cdots = \beta_m = 0$

Compute rank(**A**): Step 1. Transform into row echelon form using elementary row operations

Step 2. The number of non-zero rows in the row echelon form.

https://stattrek.com/matrix-algebra/echelon-transform#MatrixA

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Review of Fundamentals

❖ Operations on Vectors and Matrices

-**Matrix inverse**: $\mathbf{A}^{-1} = \dfrac{1}{\det(\mathbf{A})}\mathrm{adj}(\mathbf{A})$

Matrix inverse $\mathbf{A}^{-1}$ exists if and only if the matrix $\mathbf{A}$ is **square and nonsingular** (i.e., **invertible)**

e.g., $\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$ $\begin{bmatrix} 1 & 2 \end{bmatrix}$ $\begin{bmatrix} 3 & 1 \\ 9 & 3 \end{bmatrix}$

$\begin{bmatrix} 1 & 3 \\ 0 & 9 \\ 3 & 4 \end{bmatrix}$ $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 3 & 4 & 5 \end{bmatrix}$ $2 \times R_2 + R_1 = R_3$

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Review of Fundamentals

❖ Some properties on Matrix Operations

$$\mathbf{A}+\mathbf{B}=\mathbf{B}+\mathbf{A}$$

$$(\mathbf{A}+\mathbf{B})+\mathbf{C}=\mathbf{A}+(\mathbf{B}+\mathbf{C})$$

$$(\mathbf{AB})\mathbf{C}=\mathbf{A}(\mathbf{BC})$$

$$(\mathbf{A}+\mathbf{B})\mathbf{C}=\mathbf{AC}+\mathbf{BC}$$

$$\mathbf{I}_m\mathbf{A}_{m\times n}=\mathbf{A}=\mathbf{A}_{m\times n}\mathbf{I}_n$$

$$(\mathbf{A}^T)^T=\mathbf{A}$$

$$(\mathbf{A}+\mathbf{B})^T=\mathbf{A}^T+\mathbf{B}^T$$

$$(r\mathbf{A})^T=r\mathbf{A}^T$$

$$(\mathbf{AB})^T=\mathbf{B}^T\mathbf{A}^T$$

$$\mathbf{AA}^{-1}=\mathbf{A}^{-1}\mathbf{A}=\mathbf{I}$$

$$(r\mathbf{A})^{-1}=r^{-1}\mathbf{A}^{-1}$$

$$(\mathbf{AB})^{-1}=\mathbf{B}^{-1}\mathbf{A}^{-1}$$

$$\mathbf{I}^{-1}=\mathbf{I}$$

$$(\mathbf{A}^T)^{-1}=(\mathbf{A}^{-1})^T$$

$$(\mathbf{A}^{-1})^{-1}=\mathbf{A}$$

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Review of Fundamentals

❖ Sets, Functions

-A **Set** is an unordered collection of unique elements

Denoted by a calligraphic capital letter (e.g., $\mathcal{X}, \mathcal{Y}$)

$x \in \mathcal{X}$: the element x belongs to the set $\mathcal{X}$

- ✓ Finite set: a fixed number of elements

  Denoted by accolades (e.g., $\{1, 3.5, 7, -1\}$)

- ✓ Infinite set: include all values in some interval

  Denoted by square brackets as [a, b] if including a and b (e.g., [1,2])

  Denoted by parentheses as (a, b) if does not include a and b (e.g., (1,2))

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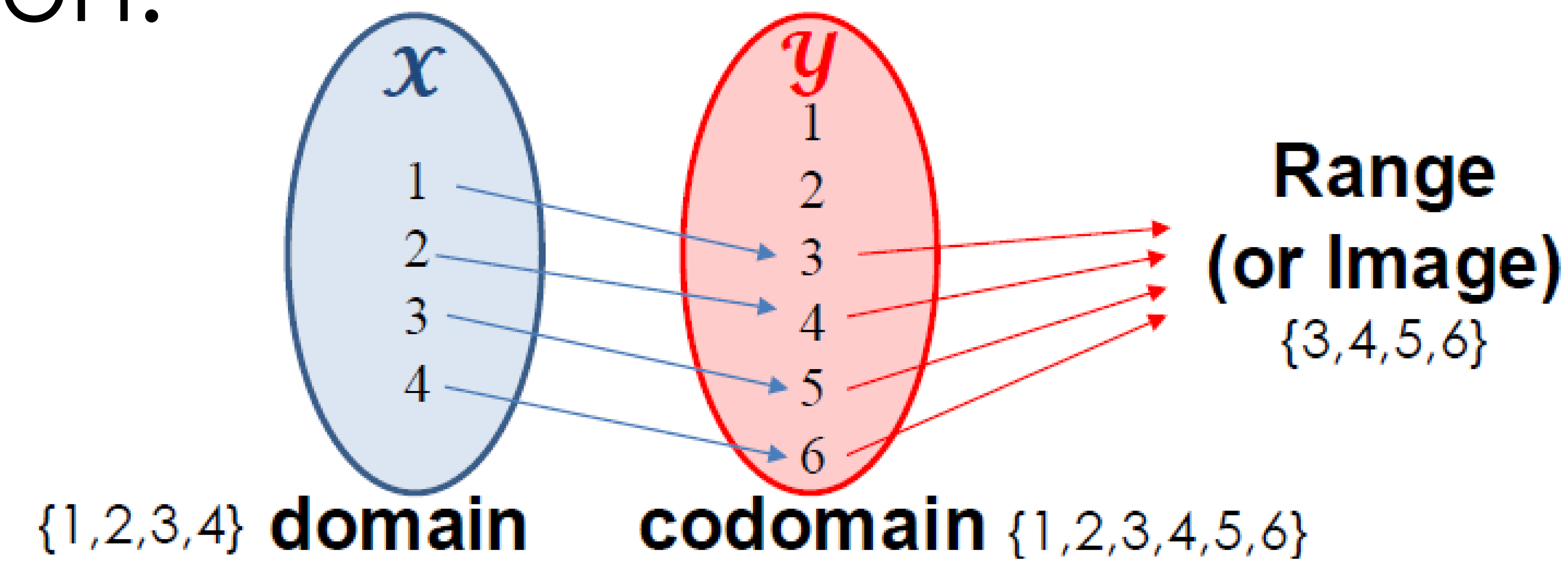
Review of Fundamentals

❖ Sets, Functions

-A **Function** is a relation that associates each element x of a set $\mathcal{X}$, the domain of the function, to a single element y of another set $\mathcal{Y}$, the codomain of the function.



Value/output

$$y = f(x)$$

Argument/input

✓ Scaler function: function returns a scalar (e.g., $f(x)$ *or* $f(\mathbf{x})$)

✓ Vector function: function returns a vector (e.g., $\mathbf{f}(x)$ *or* $\mathbf{f}(\mathbf{x})$)

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Review of Fundamentals

❖ Operations on Functions

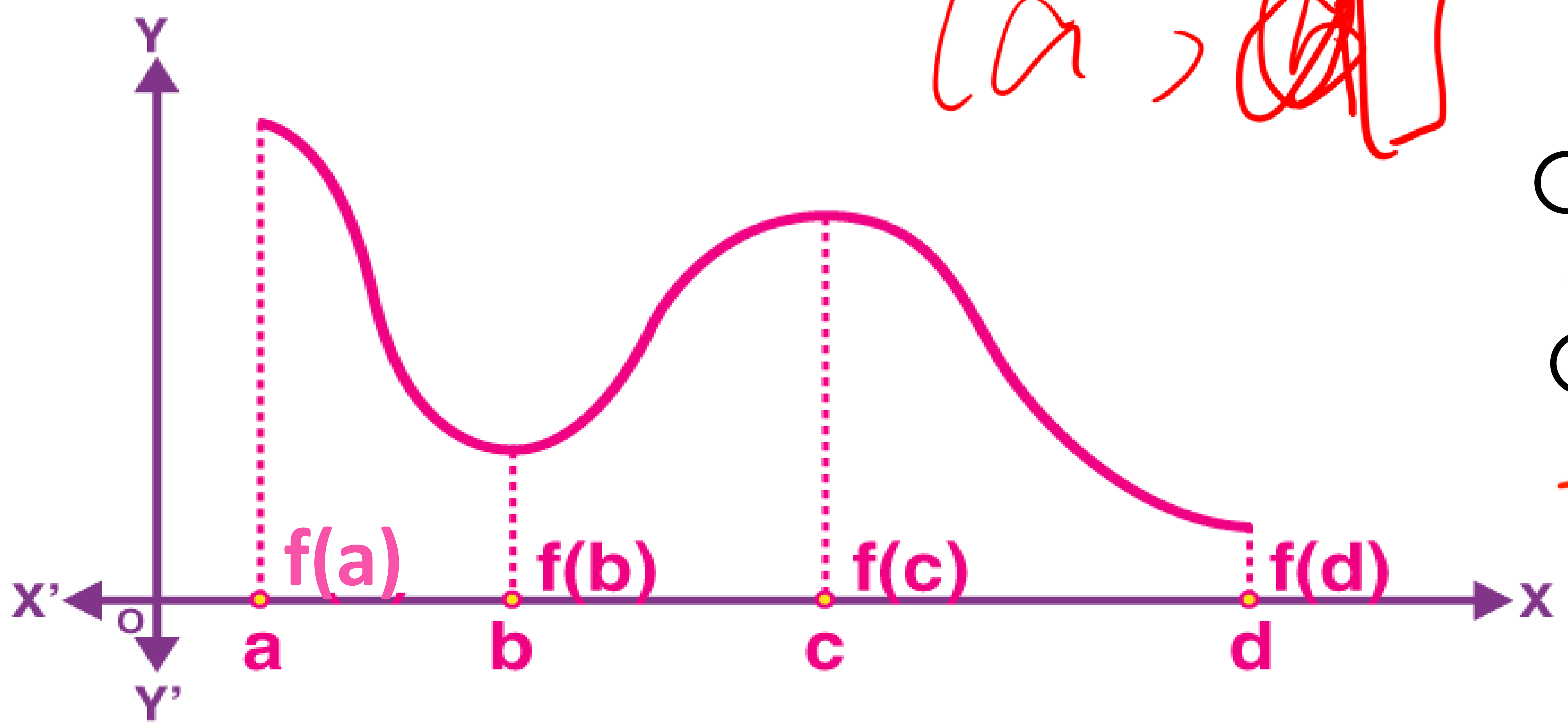
-Maximum and Minimum

$\max\limits_{x} f(x)$: returns the highest value of $f(x)$ across its domain

$\operatorname{argmax}\limits_{x} f(x)$: returns x that maximizes $f(x)$

$\min\limits_{x} f(x)$: returns the lowest value of $f(x)$ across its domain

$\operatorname{argmin}\limits_{x} f(x)$: returns x that minimizes $f(x)$



Global maximum = $\operatorname{argmax}\limits_{x} f(x)$

Global minimum = $\operatorname{argmin}\limits_{x} f(x)$

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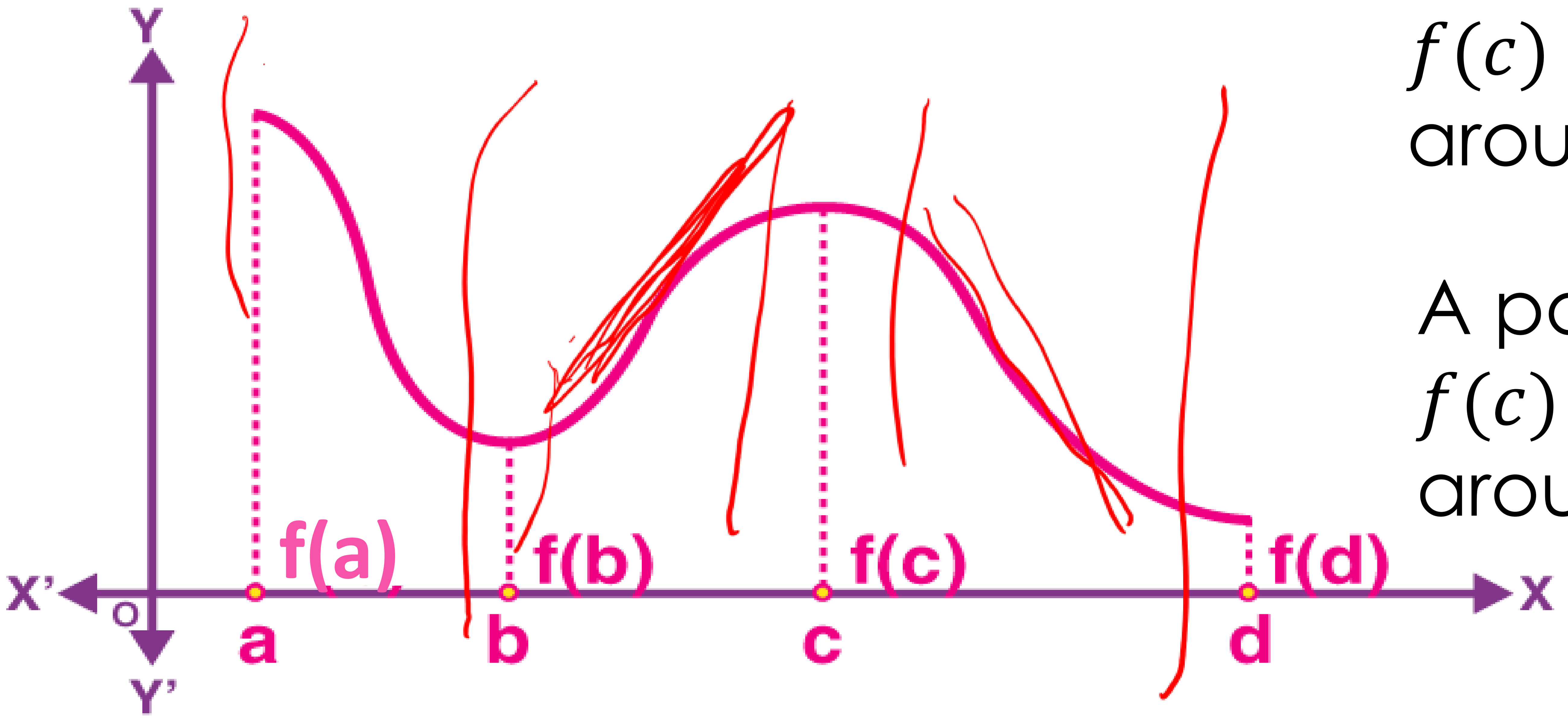
Review of Fundamentals

❖ Operations on Functions

-Derivative (single-input)

The derivative f' of a function f is a function that describes how fast f grows (or decreases)

- If f' is positive at some x, f grows at this x
- If f' is negative at some x, f decreases at this x
- If f' is zero at some x, f's slope at x is horizontal



A point x =c is a **local maximum** of $f(x)$ if $f(c) \geq f(x)$ for all x in some **open** interval around c

A point x =c is a **local minimum** of $f(x)$ if $f(c) \leq f(x)$ for all x in some **open** interval around c

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Review of Fundamentals

❖ Operations on Functions

-Gradient (multiple-input scalar function)

Generalization of derivative for scalar functions with multiple inputs.

Denoted by $\nabla_{\mathbf{x}} f$

A vector of **partial derivatives**

$$\nabla_{\mathbf{x}} f = \frac{df(\mathbf{x})}{d\mathbf{x}} = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_d} \end{bmatrix}$$

Differentiation of $f(\mathbf{x})$ w.r.t. $\mathbf{x}$

A matrix of **partial derivatives**

$$\nabla_{\mathbf{X}} f = \frac{df(\mathbf{X})}{d\mathbf{X}} = \begin{bmatrix} \frac{\partial f}{\partial x_{1,1}} & \cdots & \frac{\partial f}{\partial x_{1,K}} \\ \vdots & \ddots & \vdots \\ \frac{\partial f}{\partial x_{d,1}} & \cdots & \frac{\partial f}{\partial x_{d,K}} \end{bmatrix}$$

same size as **X**

e.g., $f(\mathbf{x}) = 2x_1 + 3x_2$, what is $\nabla_{\mathbf{x}} f$?

$\begin{bmatrix} 2 \\ 3 \end{bmatrix}$

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Review of Fundamentals

❖ Operations on Functions

-Jacobian (multiple-input vector function)

Generalization of derivative for vector functions with multiple inputs.

A matrix of **partial derivatives**

$$\frac{\partial(2x_1+3x_2)}{x_2}$$

$$\begin{bmatrix} 2 & 3 \\ x_2 & x_1 \end{bmatrix}$$

$$\frac{d\mathbf{f}(\mathbf{x})}{d\mathbf{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_h}{\partial x_1} & \cdots & \frac{\partial f_h}{\partial x_d} \end{bmatrix}$$

$$\frac{\partial(5x_1x_2)}{x_1}$$

Differentiation of $\mathbf{f}(\mathbf{x})$ of size hx1 w.r.t. $\mathbf{x}$ of size dx1.

e.g., $\mathbf{f}(\mathbf{x}) = \begin{bmatrix} 2x_1 + 3x_2 \\ 5x_1x_2 \end{bmatrix}$, what is the Jacobian matrix $\frac{d\mathbf{f}(\mathbf{x})}{d\mathbf{x}}$?

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Review of Fundamentals

❖ Common Differentiation Rules

- Constant Rule: the derivative of a constant is 0
- Sum & Difference Rule: $\left(f(x)+g(x)\right)' = f'(x)+g'(x)$

  $\left(f(x)-g(x)\right)' = f'(x)-g'(x)$
- Product Rule: $\left(f(x)g(x)\right)' = f(x)g'(x)+f'(x)g(x)$
- Quotient Rule: $\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x)-f(x)g'(x)}{(g(x))^2}$
- Chain Rule: $(f(g(x)))' = f'(g(x))g'(x)$ or $\frac{df}{dx} = \frac{df}{dg}\frac{dg}{dx}$

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